

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech.Winter 2018 - 19 Examination

Semester: 3
Subject Code: 03105202
Subject Name: Operating System

Date: 12/12/2018
Time: 2:00pm to 4:30pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (Each of one mark) (15)

1. Operating system is a
 - a) System software
 - b) Application software
 - c) Presentation software
 - d) Database software
2. In producer-consumer problem, when buffer status is partially empty _____ has to wait
 - a) Producer
 - b) Consumer
 - c) None
 - d) Both
3. If the resources are always preempted from the same process, _____ can occur
 - a) deadlock
 - b) Aging
 - c) System crash
 - d) Starvation
4. What is compaction?
 - a) a technique for overcoming internal fragmentation
 - b) a paging technique
 - c) a technique for overcoming external fragmentation
 - d) a technique for overcoming fatal error
5. Operating System maintains the PCB for:
 - a) each process
 - b) each thread
 - c) each instruction
 - d) each address
6. Differentiate between Compiler and Interpreter .
7. List different job scheduling algorithms in operating systems.
8. What is a Real-Time System?
9. Difference between Primary storage and secondary storage.
10. Define semaphore and what are its uses?
11. A system call to create a child process is _____
12. what is kernel mode?
13. Suppose that process is in BLOCK state and waiting for some I/O services when service is completed, it goes to the _____ state.
14. _____ is called lightweight process.
15. High page fault leads to _____

Q.2 Answer the following questions. (Attempt any three) (15)

- A) Write different Operating System Services with diagram.
- B) Define process. Differentiate between a process and a program. Explain different states of a process with a suitable diagram.
- C) What is Semaphore? List out different types of Semaphore. Explain Produce-Consumer Problem.
- D) What is Deadlock? List the conditions that lead to deadlock. Explain any two ways to get recovery from Deadlock.

Q.3 A) Explain the following in detail with suitable diagrams: (i) Swapping (ii) Segmentation. (07)

- (iii) Multiprogramming with fixed partitions (iv) Multiprogramming with Variable Partitions

B) What is called TLB? How does it help to speed up paging? (08)

OR

B) Explain Context Switching. Discuss performance evaluation of FCFS (First Come First Serve) & RR (Round Robin) scheduling. **(08)**

Q.4 A) Explain Bankers algorithm in detail.. **(07)**

OR

A) Suppose Disk drive has 200 cylinders. The current position of head is 53. The queue of pending request is 98, 183, 37, 122, 14, 124, 65, 67. The Head pointer is at 53. Calculate head movement for the following algorithms. **(07)**

(i) FCFS

(ii) SSTF

B) For the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 **(08)**
Calculate the page faults applying the following Page Replacement Algorithms for a memory with three frames: (i) FIFO (ii) Optimal (iii) LRU .Explain the three techniques